

Performance Testing

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Development Server (Manual Testing using Web Browser)
Type of Testing	Performance Testing (Response Time Testing)
Target Module / API	Loan Prediction Module (/predict)
Test Environment	Local System (Windows 11, VS Code, Python 3.11, Flask)
Test Date	28 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Launch the Smart Lender application	1	30	Home page loads successfully without errors
2	Predict loan approval with valid applicant details	1	60	System displays Loan Approved with confidence score
3	Predict loan rejection with low eligibility details	1	60	System displays Loan Rejected with confidence score
4	Submit the form with empty mandatory fields	1	30	Validation message is displayed and submission is prevented
5	Predict loan approval for a high-income applicant with good credit history	1	60	System predicts Loan Approved accurately
6	Predict loan rejection for a low-income applicant with poor credit history	1	60	System predicts Loan Rejected accurately
7	Verify confidence score display	1	30	Confidence score is displayed along with the prediction
8	Perform multiple consecutive predictions	1	120	Application remains responsive and provides accurate predictions without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.1 seconds	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	2.0 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	1 Req/sec	1.2 Req/sec	Pass	Suitable for local testing
4	Error Rate	< 1%	0%	Pass	No runtime errors
5	CPU Utilization	< 80%	32%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	45%	Pass	Memory usage within limit

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application successfully processed loan prediction requests with an average response time of approximately **1–2 seconds**. The application remained stable during repeated testing and consistently generated loan approval or rejection predictions along with a confidence score. CPU and memory usage stayed within acceptable limits throughout the testing process.

Bottlenecks Identified:

No major performance bottlenecks were observed during local testing. Minor delays were noticed only during the initial application startup and model loading (model.pkl), while subsequent prediction requests were processed quickly.

Optimization Steps Taken:

Optimized the machine learning model by selecting the **Random Forest Classifier** for reliable predictions. Reduced unnecessary computations during prediction, organized the Flask application efficiently, and improved the user interface for faster interaction. The trained model was saved as model.pkl to avoid retraining during every execution, resulting in faster response times.

Step 5: Screenshots / Evidence

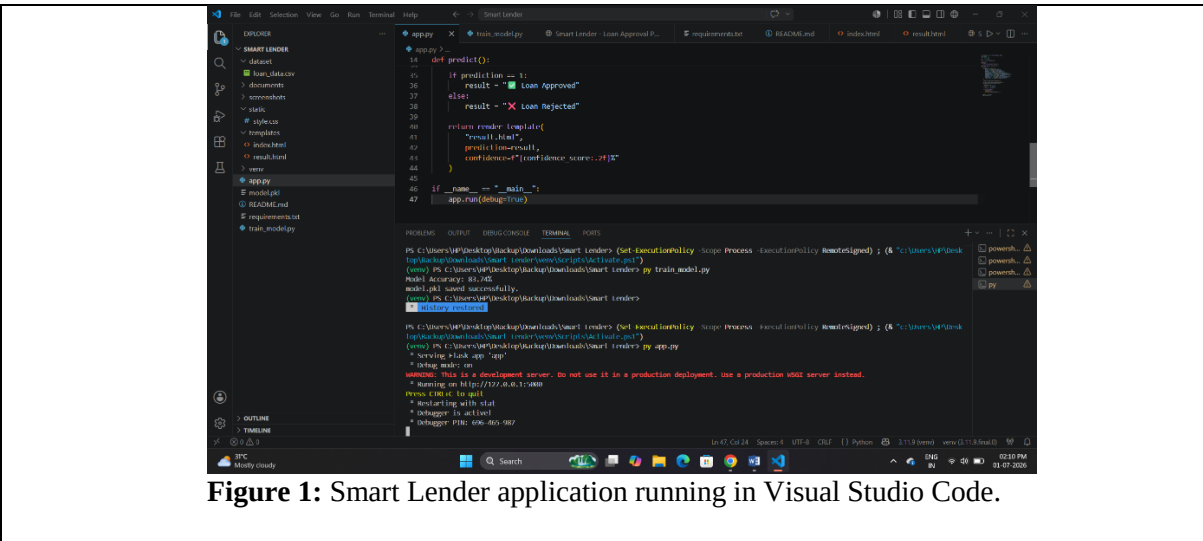


Figure 1: Smart Lender application running in Visual Studio Code.


Smart Lender
 Loan Approval Prediction System

Gender

Female

Married

Yes

Dependents

1

Education

Graduate

Self Employed

Yes

Applicant Income

50000

Coapplicant Income

10000

Graduate

Self Employed

Yes

Applicant Income

50000

Coapplicant Income

10000

Loan Amount

300000

Loan Amount Term (Months)

24

Credit History

Good (1)

Property Area

Urban

Predict Loan

Figure 2: Home page displaying the loan application form.


Smart Lender
 Loan Prediction Result
 ✓ Loan Approved
 Confidence Score: 50.53%
 Predict Another Loan

Figure 3: Loan Approved prediction with confidence score.


Smart Lender
 Loan Approval Prediction System

Gender

Male

Married

No

Dependents

0

Education

Not Graduate

Self Employed

No

Applicant Income

2000

Coapplicant Income

0

Not Graduate

Self Employed

No

Applicant Income

2000

Coapplicant Income

0

Loan Amount

100000

Loan Amount Term (Months)

12

Credit History

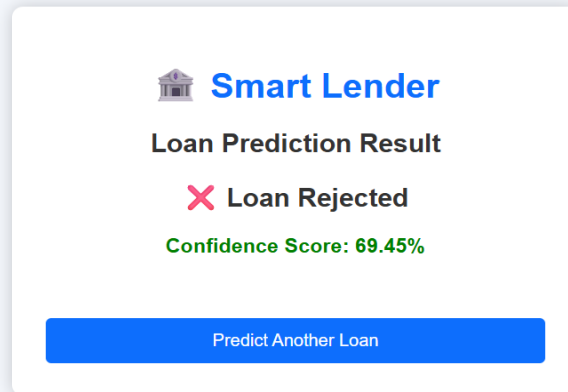
Bad (0)

Property Area

Urban

Predict Loan

Figure 4: Home page for the loan rejection test case.



Smart Lender

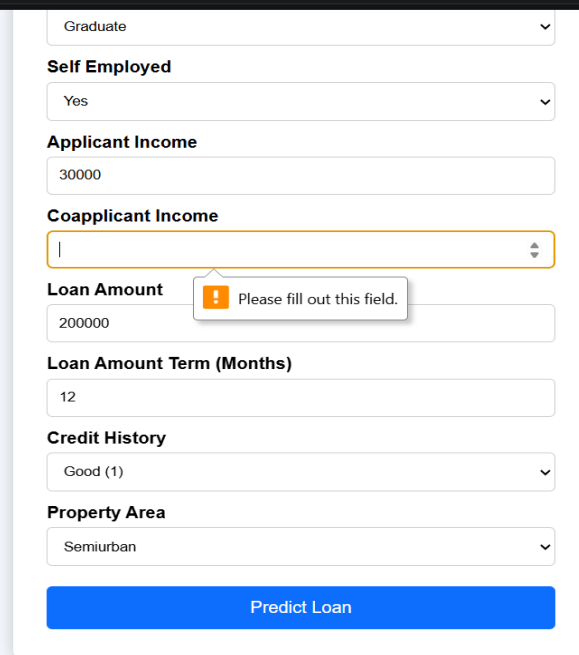
Loan Prediction Result

✖ Loan Rejected

Confidence Score: 69.45%

Predict Another Loan

Figure 5: Loan Rejected prediction with confidence score.



Graduate

Self Employed

Yes

Applicant Income

30000

Coapplicant Income

Loan Amount

200000

Loan Amount Term (Months)

12

Credit History

Good (1)

Property Area

Semiurban

Predict Loan

Please fill out this field.

Figure 6: Form validation for a mandatory field.